Lecture - 22

14/11/2025

Matrix of a Linear Transformation

Let $T: V \to W$ be a linear transformation of finite dimensional vector spaces. Let $\mathcal{B}_1 = \{v_1, \dots, v_n\}$ and $\mathcal{B}_2 = \{w_1, \dots, w_m\}$ be bases of V and W respectively.

Then,

$$
\begin{aligned}
T(v_1) &= a_{11} w_1 + a_{21} w_2 + \dots + a_{m1} w_m \\
T(v_2) &= a_{12} w_1 + a_{22} w_2 + \dots + a_{m2} w_m \\
&\vdots \\
T(v_n) &= a_{1n} w_1 + a_{2n} w_2 + \dots + a_{mn} w_m
\end{aligned}
$$

The matrix

$$
M = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}
$$

is called the matrix of the linear transformation T with respect to the basis $\mathcal{B}_1$ & $\mathcal{B}_2$.

Observation:

① Let $T: \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be a linear transformation and consider the standard basis $e_1, e_2, \dots, e_n$ of $\mathbb{R}^n$

unit vector basis $e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \end{pmatrix}$, $e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}$, $\dots$

Let

$$T(e_i) = a_{1i}e_1 + a_{2i}e_2 + \cdots + a_{ni}e_i = \begin{pmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{ni} \end{pmatrix}$$

for each $1 \leq i \leq n$.

Then $T(e_i) = \mathcal{M}e_i$,

where $\mathcal{M}$ is the matrix of the linear transformation with respect to the standard basis.

Furthermore,

$$\begin{aligned} T(v) = T(\alpha_1 e_1 + \alpha_2 e_2 + \cdots + \alpha_n e_n) &= \alpha_1 T(e_1) + \cdots + \alpha_n T(e_n) \\ &= \alpha_1 \mathcal{M} e_1 + \cdots + \alpha_n \mathcal{M} e_n \\ &= \mathcal{M}(\alpha_1 e_1 + \cdots + \alpha_n e_n) \\ &= \mathcal{M} v. \end{aligned}$$

$= \mathcal{M}v$ for any $v \in \mathbb{R}^n$.

Thus every linear transformation $T: \mathbb{R}^n \to \mathbb{R}^n$ is simply a matrix multiplication.

(2) Let $T: V \longrightarrow W$ be a linear transformation, $\mathcal{B}_1 = \{v_1, \ldots, v_n\}$ and $\mathcal{B}_2 = \{w_1, \ldots, w_m\}$ be bases of V and W respectively and

$$\mathcal{M} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$ be the matrix of

T with respect to $\mathcal{B}_1$ & $\mathcal{B}_2$.

Then, from the definition,

$$T(v_i) = a_{1i} w_1 + a_{2i} w_2 + \cdots + a_{mi} w_m .$$

If $v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$ then

$$\begin{aligned}
T(v) &= \alpha_1 T(v_1) + \cdots + \alpha_n T(v_n) \\
&= \alpha_1 (a_{11} w_1 + a_{21} w_2 + \cdots + a_{m1} w_m) + \cdots + \alpha_n (a_{1n} w_1 + \cdots + a_{mn} w_m) \\
&= (\alpha_1 a_{11} + \alpha_2 a_{12} + \cdots + \alpha_n a_{1n}) w_1 + (\alpha_1 a_{21} + \alpha_2 a_{22} + \cdots + \alpha_n a_{2n}) w_2 \\
&\quad + \cdots + (\alpha_1 a_{m1} + \alpha_2 a_{m2} + \cdots + \alpha_n a_{mn}) w_n
\end{aligned}$$

Note that

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} \alpha_1 a_{11} + \alpha_2 a_{12} + \cdots + \alpha_n a_{1n} \\ \alpha_1 a_{21} + \alpha_2 a_{22} + \cdots + \alpha_n a_{2n} \\ \vdots \\ \alpha_1 a_{m1} + \alpha_2 a_{m2} + \cdots + \alpha_n a_{mn} \end{pmatrix}$$

Thus if $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ is thought of as the vector $\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}$ then $T(v)$ can be thought of as the vector

$$\begin{pmatrix} \alpha_1 a_{11} + \alpha_2 a_{12} + \cdots + \alpha_n a_{1n} \\ \alpha_1 a_{21} + \alpha_2 a_{22} + \cdots + \alpha_n a_{2n} \\ \vdots \\ \alpha_1 a_{m1} + \alpha_2 a_{m2} + \cdots + \alpha_n a_{mn} \end{pmatrix}$$

$$\updownarrow$$

$$(\alpha_1 a_{11} + \alpha_2 a_{12} + \cdots + \alpha_n a_{1n}) w_1 + (\alpha_1 a_{21} + \alpha_2 a_{22} + \cdots + \alpha_n a_{2n}) w_2 + \cdots$$
$$+ (\alpha_1 a_{m1} + \alpha_2 a_{m2} + \cdots + \alpha_n a_{mn}) w_m$$

Example

Let $T: V \to W$ be a linear transformation of finite dimensional vector spaces; $\mathcal{B}_1 = \{v_1, v_2, v_3\}$ and $\mathcal{B}_2 = \{w_1, w_2\}$ be bases of V and W respectively.

Let

$$M = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 0 & 3 \end{pmatrix}$$ be the matrix of T

with respect to $\mathcal{B}_1$ and $\mathcal{B}_2$. Calculate $T(v_1 + 2v_2 + 3v_3)$.

Solution Think of $v_1 + 2v_2 + 3v_3$ as $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

Since $\begin{pmatrix} 1 & 3 & 1 \\ 2 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 10 \\ 11 \end{pmatrix}$, we obtain that

$$T(v_1 + 2v_2 + 3v_3) = 10w_1 + 11w_2.$$

Eigenvalues, eigenvectors of a square matrix

Let A be an $n \times n$ matrix. A number $\lambda \in \mathbb{R}$ is called an eigenvalue if there is a nonzero vector $v \in \mathbb{R}^n$, called an eigenvector,

such that $Av = \lambda v$.

Proposition. Let A be an $n\times n$ matrix. Then A has an eigenvector if and only if there is a line L in $\mathbb{R}^n$ such that $AL = L$;

Proof Let $v \neq 0$ be an eigenvector of A for an eigenvalue λ. Then, $Av = \lambda v$ and therefore

$$A(\alpha v) = \alpha A v = \alpha \lambda v = \lambda \alpha v.$$

This implies that if $L = \{\alpha v : \alpha \in F\}$ then

$$AL = \{\alpha A v : \alpha \in F\} = \{\lambda \alpha v : \alpha \in F\}$$
$$= \{\beta v : \beta \in F\} = L.$$

Conversely, if $AL = L$ for some line L, then pick a nonzero vector $v \in L$ and observe that

$$L = \{\alpha v : \alpha \in F\}.$$

Now $AL = L \Rightarrow Av \in L$. That is, $Av = \lambda v$ for some $\lambda \in F$. Thus, v is an eigenvector of A.

Method of finding eigenvalues & eigenvectors of an $n\times n$ matrix

Let A be an $n\times n$ matrix. We need to find all $\lambda \in \mathbb{R}$ so that $Av = \lambda v$ for some $v \neq 0$.

That is, find all $v \neq 0$ so that $(A - \lambda I_n)v = Av - \lambda I_n v$

$$= Av - \lambda v$$
$$= 0$$

Thus we need to Calculate $\ker(A-\lambda I_n)$. Note that $A-\lambda I_n$ is a matrix and therefore there is a nonzero v such that $(A-\lambda I_n)v = 0$ if and only if $\det(A-\lambda I_n)=0$

Therefore;

1. Calculate $\det(A-\lambda I_n)=0$. You will get a polynomial in λ.

2. Find all roots of the polynomial in $\mathbb{R}$, if it exist. Those are the eigenvalue.

3. For those eigenvalues λ, $\ker(A-\lambda I_n) \neq \{0\}$. Thus any $0 \neq v \in \ker(A-\lambda I_n)$ is an eigenvector

Example / non examples

1. Find all eigenvectors and eigenvalues of $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = A$.

We need to find all $\lambda \in \mathbb{R}$ such that $Av = \lambda v$.

Now $\det(A-\lambda I_n) = \det\left(\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right)$

$= \det\begin{pmatrix} 1-\lambda & 2 \\ 0 & 3-\lambda \end{pmatrix}$

$$= (1-\lambda)(3-\lambda)$$

Then, $\det(A-\lambda I_n) = 0$ for $\lambda = 1$ and $\lambda = 3$.

<u>Eigenvalues</u> : $\lambda = 1$, $\lambda = 3$.

<u>Eigenvectors</u>: $\boxed{\text{Let } \lambda = 1}$. Then $A - \lambda I_n = \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix}$.

Any nonzero vector $\begin{pmatrix} x \\ y \end{pmatrix} \in \operatorname{Ker}(A-\lambda I_n)$ is an eigenvector.

$$\text{Since, } \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$\Rightarrow 2y = 0 \Rightarrow y = 0.$$

We see that any vector of the form $\begin{pmatrix} x \\ 0 \end{pmatrix}$, where $x \neq 0$, is an eigenvector.

In particular, $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \ker \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix}$ is an eigenvector of A for the eigenvalue $\lambda = 1$.

$\boxed{\text{Let } \lambda = 3}$ Then, we need to find the kernal of

$$\begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix}.$$

Let $\begin{pmatrix} x \\ y \end{pmatrix} \in \ker \begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix}$. Then $\begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$

$$\Rightarrow -2x + 2y = 0 \Rightarrow x = y.$$

Thus an nonzero vector of the form $\begin{pmatrix} x \\ x \end{pmatrix}$ is an eigenvector of

$$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}. \quad \text{for the eigenvalue } \lambda = 3$$

In particular, $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector of $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ for the eigenvalue $\lambda = 3$.

Geometrically, the line $\underbrace{\left\{ \begin{pmatrix} x \\ 0 \end{pmatrix} : x \in \mathbb{R} \right\}}_{x\text{-axis}}$ and

the line $\left\{ \begin{pmatrix} x \\ x \end{pmatrix} : x \in \mathbb{R} \right\}$ are "invariant" under the transformation $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$

Eigenvector Basis

We Consider example 1.

$$A: \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \quad A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$$

Note that
$$A e_1 = 1 \cdot e_1 + 0 e_2$$
$$A e_2 = 2e_1 + 3e_2$$

So the matrix of linear transformation is A, with respect to the Basis $\{e_1, e_2\}$.

We know that